Simulating Climate Risk Migration

The Shadow Banking Spillover Effect

Vaibhav Thakur

BS Economics, IISER Bhopal

Outline

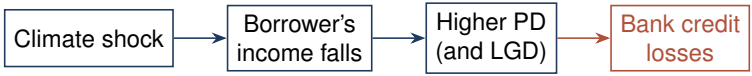
1. Climate risk in banking
2. Regulation and disclosure
3. Research question
4. Model design
5. Mechanism
6. Worked example
7. Results
8. Conclusion

The one question

When a bank prices climate risk *correctly*, does the risk disappear—or does it move somewhere else?

Why should a bank care about climate?

A bank does not lend against the weather; it lends to *borrowers*. **Climate risk is credit risk.**



	Physical risk	**Transition risk**
Source	Floods, drought	Carbon tax, coal phase-out
Hits	Farms, coasts	Power, steel, cement
Channel	Income, collateral	Profit, stranded assets

Exposures

Corporate loans
Home loans in flood zones
MSME and farm loans
Bond and equity holdings

Measuring the damage: expected loss

Lend 100, ask three questions

- **PD**: chance of not being repaid. 3 in 100 means PD = 3%.
- **LGD**: share lost after recoveries. Recover 40 and LGD = 60%.
- **EAD**: amount still owed at default. Here, 100.

$$EL = PD \times LGD \times EAD$$

A shock raises PD, and sometimes LGD

Event	EAD	PD	LGD	EL
Flood hits a factory	500	3%→15%	60%	9→45
Flood-zone mortgages	15,000	2%→5%	30%→50%	90→375
Carbon policy, one sector	20,000	4%→8%	50%	400→800

In the mortgage row both move: borrowers pay less often and the houses backing the loans are worth less. EL rises more than fourfold.

From a loss to a thinner capital cushion



- **CET1**: highest-quality capital, mainly equity and retained earnings. The first line of defence.
- **RWA**: assets weighted by riskiness; riskier loans count for more.
- **CAR**: capital divided by RWA. Regulators set a minimum.

The double squeeze (Rs crore; CAR 11.5%)

	Before	After
Capital	575	475
RWA (weight 100%→120%)	5,000	6,000
Capital ratio	11.5%	7.9%

A Rs 100 crore loss pushes capital down while the downgrade pushes RWA up.

Why it becomes a system-wide problem

- **Concentration.** Banks lend to the same sectors, so one policy change can hurt every bank at once.
- **Amplification.** Each sector's loss looks small. Repeated across sectors, the losses add up.
- **Credit tightening.** Banks under pressure charge more and lend less. Firms that lose credit become weaker, which creates more losses: climate stress is *procyclical*.

> **The bridge to this project.** If a bank cuts credit to weaker borrowers to protect itself, those borrowers do not disappear. They look for money elsewhere.

How do regulators test for climate risk?

NGFS: a voluntary group of central banks and supervisors (since 2017). No binding rules, but common scenarios that regulators adapt.

Scenario	What it looks like
Orderly	Early action; small shock
Disorderly	Late, sudden carbon-price spike
Hot house world	Little action; severe physical damage

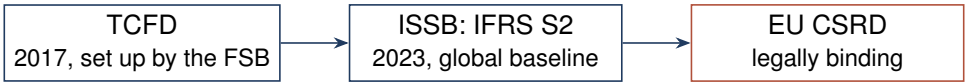
Two tools, two questions

- **Stress test**: can the bank survive the scenario?
- **Climate VaR**: how much could be lost?

Challenge: scenarios run 20–30 years, supervisory stress tests look 1–3 years ahead.

Example. Rs 10,000 crore lent to thermal power; a carbon tax lifts PD from 2% to 6% (LGD 50%). EL rises from 100 to 300 and the CET1 ratio falls from 12% to 11.87%: mild alone, but it compounds across sectors.

Disclosure: how do markets learn about climate risk?



A voluntary framework with four pillars: governance, strategy, risk management, metrics and targets. It became the starting point for everything after it.

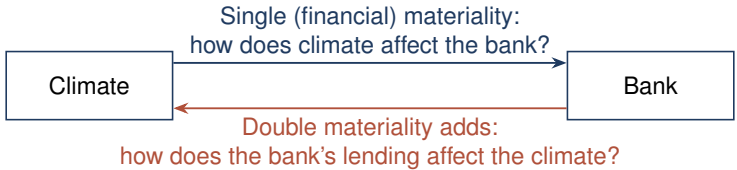
Builds the TCFD pillars into a standard and adds Scope 1, 2 and 3 emissions and scenario analysis. The TCFD disbanded in 2023 as its work moved to the ISSB.

Makes such reporting a legal duty in the EU, and is built on *double* materiality (next slide).

> Without disclosure, no one outside the bank can price or supervise its climate risk.

Single and double materiality

Something is *material* if it is important enough to change decisions of investors or regulators.



Example. A bank finances 20 coal projects, each emitting 1 million tons of CO_2 a year, so its lending is linked to 20 million tons. A transition policy that raises the PD of those loans is also a financial risk to the bank. The bank is exposed to climate risk *and* a contributor to it.

Measuring a bank's footprint: financed emissions and taxonomies

Financed emissions

A bank's own offices (Scope 1 and 2) are small. Most of its footprint is Scope 3: the firms it finances.

Steel firm emits 5 Mt a year; the bank funds 20% of it.
Financed emissions = $0.20 \times 5 =$ **1 Mt**

Taxonomy and Green Asset Ratio

A taxonomy is a rulebook for what counts as **green** (solar, wind), **transition** (gas, under conditions) or **brown** (coal mining).

$$\text{GAR} = \frac{\text{taxonomy-aligned assets}}{\text{total eligible assets}}$$

EUR 100 bn of 500 bn: **20%**

The Indian context

SEBI: BRSR	Business Responsibility and Sustainability Reporting: mandatory from FY 2022–23 for the top 1,000 listed firms (emissions, energy use, ESG risks).
RBI, 2022	Discussion paper on climate risk and sustainable finance; climate risk treated as a financial-stability concern.
RBI, 2024 draft	Disclosure framework on the four TCFD pillars for banks and large NBFCs. Proposed phase-in: governance, strategy, risk management from FY 2025–26; metrics and targets from FY 2027–28.
Basel link	In line with the Basel Committee's 2022 principles; includes capital assessment (ICAAP).

The RBI framework is shown as proposed in the February 2024 draft; check the RBI website for the final text.

What happens to a borrower the bank turns away?

What we already know (India)

- After local floods, bank branches raise rates (Das, Majilla & Saurav).
- Non-banks (NBFCs) expand lending in the same areas (Mathur & Sarma).

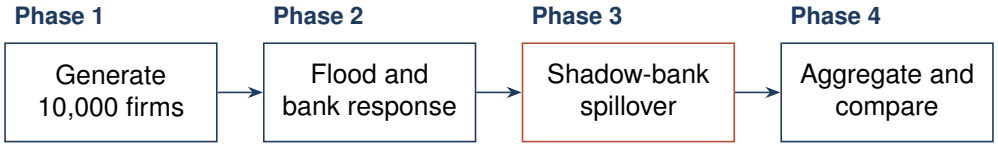
NBFC: lends like a bank, takes no ordinary deposits, lighter regulation. With similar lenders it forms the "shadow banking" sector.

Two testable implications

1. The bank's remaining loan book looks *safer* after the shock.
2. The firms pushed out default at a *far higher* rate.

> **Hypothesis.** Strict climate regulation of banks does not eliminate systemic climate risk; it moves it into the opaque shadow-banking sector.

A synthetic economy, built in four phases



Why a simulation?

No public dataset links, for the same firm, its flood exposure, its bank's decision and its later NBFC borrowing.

Why synthetic data?

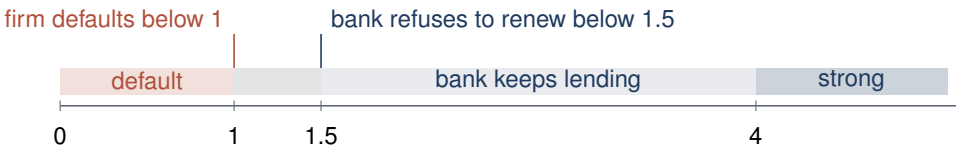
It isolates the mechanism. Firms are calibrated to published RBI data on firm health, so the starting point is not arbitrary.

> This is a **proof of mechanism**, not a forecast for the Indian economy.

Repayment capacity in one number: the ICR

$$ICR = \frac{\text{EBITDA}}{\text{interest expense}}$$

"How many times does operating profit cover the interest bill?"



- A flood lowers the numerator (income). Lenders' response raises the denominator (interest).
- ICR below 1.5 is widely read as acute financial stress.

Phase 1: build 10,000 firms, anchored to RBI data

Each firm holds one bank loan at 8.5% before the flood.

- Revenue: normal(500, 150), clipped to 100–2000
- EBITDA margin: uniform, 10–30%
- Debt = multiple of revenue: 0.1–0.6× (50% of firms), 0.6–2.5× (25%), 2.5–4.5× (25%)
- Flood vulnerability v: integer 1–100, independent of finances

Is the baseline realistic?

Baseline ICR	Model	RBI
Strong (> 4)	43.8%	47.6%
Moderate (1–4)	29.0%	30.5%
Vulnerable (≤ 1)	27.2%	21.9%

Close to the RBI pattern, with slightly more vulnerable firms.

RBI: listed non-financial firms, H1 2023–24 (report of 28 Dec 2023).

Phase 2: the flood hits income; banks reprice and ration

1. **Shock.** Revenue falls by 35% $\times$ $v/100$. A firm with $v = 100$ loses 35%; margins are held fixed, so EBITDA falls in proportion.
2. **Reprice.** Banks raise the rate from 8.5% to 9.0% for firms with $v > 50$.
3. **Ration.** Banks renew a loan only if the stressed ICR is at least 1.5.

Outcome in the simulation

Aggregate EBITDA	–17.6%
Loans renewed	5,896
Loans refused	4,104 (41%)

About 90% of refused firms already had an ICR below 1.5 before the flood: the shock tips weak firms over the edge.

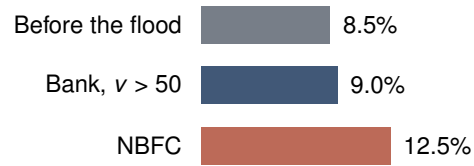
> Banks here do exactly what regulators want: they refuse firms that have become too risky.

Phase 3: refused firms turn to NBFCs

- **Full absorption.** NBFCs lend to every refused firm.
- **Risk premium.** They add a 400 bps premium: 12.5%, up from 8.5%.
- **Default.** A firm defaults if its final ICR is below 1: profit cannot even pay the interest.

Default here means financial distress (interest not covered), not a legal bankruptcy.

Interest rate each firm pays



> The 400 bps premium is illustrative, and it is the most influential parameter of this phase.

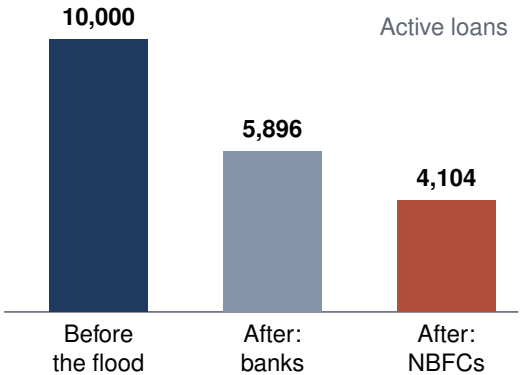
One firm, end to end

A firm with EBITDA of 100, debt of 600 and flood vulnerability $v = 80$ (revenue and EBITDA fall by $0.35 \times 0.80 = 28\%$). Interest = debt $\times$ rate; ICR = EBITDA / interest.

Stage	EBITDA	Rate	Interest	ICR	Outcome
Before the flood	100	8.5%	51	1.96	healthy
After flood, bank rate	72	9.0%	54	1.33	refused (below 1.5)
After flood, NBFC rate	72	12.5%	75	0.96	default (below 1)

> The firm is the same and its debt is the same. Only the shock and the lender changed, and the firm moved from healthy to distressed.

Result 1: credit migrates, and the bank book looks safer

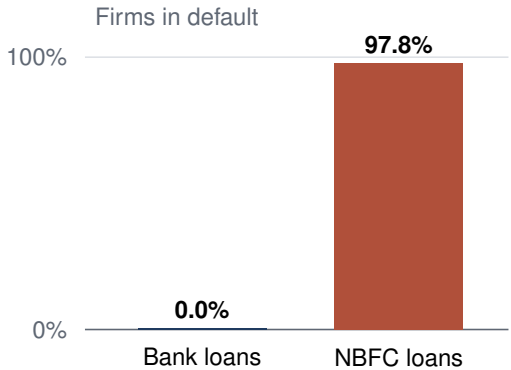


41% of loans move to NBFCs, but they carry **81%** of all debt: the firms that move are the most heavily indebted.

Among firms that stay with banks, the share with a strong pre-shock ICR rises from **44%** to **74%**.

A stress test that looks only at banks would see a stronger portfolio after the flood.

Result 2: the migrated risk is fragile



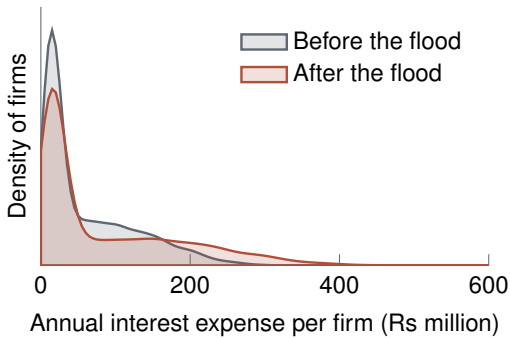
0 of 5,896 bank borrowers default.

4,013 of 4,104 NBFC borrowers default (97.8%); only 91 survive.

All defaults sit in a portfolio that bank-level statistics do not show.

Partly by construction: these firms were already weak and the premium is large. See the limitations.

Result 3: the economy-wide interest burden rises



Total interest paid by the economy rises by **39%**.

Two drivers:

- a higher bank rate for exposed firms
- heavily indebted firms moving to a lender that charges 12.5%

The right tail of large interest bills becomes thicker.

Conclusion: what we learned, and how far to trust it

What we learned

1. Correct pricing of climate risk by banks does not remove the risk from the economy.
2. The risk migrates to lighter-regulated lenders, where it is harder to see.
3. Bank-only stress tests can look reassuring while systemic risk grows.

How far to trust it

- Synthetic firms: a proof of mechanism, not a forecast.
- One-off, linear shock; no collateral or recovery.
- Full NBFC absorption and the 400 bps premium are assumed.
- Default means ICR below 1, not bankruptcy.

> **Next step:** vary the premium and the 1.5 cutoff to test how sensitive the results are, then use real lender-level data.

References

Das, A., Majilla, T., & Saurav, S. *Natural Disasters, Interest Rate Dynamics, and Economic Activities.* CAFRAL.

Mathur, S., & Sarma, R. *Who Lends When It Floods? Banks vs Non-Banks.* CAFRAL.

Reserve Bank of India (2023). Report, 28 December 2023 (interest coverage ratio of listed non-financial firms).

Basel Committee on Banking Supervision (2022). *Principles for the Effective Management and Supervision of Climate-Related Financial Risks.* BIS.

Network for Greening the Financial System (2023). *NGFS Climate Scenarios for Central Banks and Supervisors.*

Task Force on Climate-related Financial Disclosures (2017). *Final Report: Recommendations of the TCFD.* Financial Stability Board.

Thank you

Questions and discussion